Tasnimul Hasan



Computer Science & Engineering Undergraduate

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PROFESSIONAL SUMMARY

Driven Computer Science & Engineering undergraduate with a strong foundation in Object-Oriented Programming, Data Structures & Algorithms, and embedded systems development. Demonstrated aptitude in designing functional lab prototypes, coordinating collaborative technical projects, and applying systematic problem-solving methods. Seeking a software engineering internship or trainee position to deliver value through rigorous coding and continuous learning.

TECHNICAL PROFICIENCY

Programming:	Java, C, Python (Basic)
Core CS:	Data Structures (Arrays, Linked Lists, Stacks, Queues, Trees, BST, Heaps, Graphs, Hashing), Algorithms, OOP (Inheritance, Abstraction, Encapsulation)
Embedded Systems:	ESP32 DevKit V1, TCRT5000 IR Sensor Arrays, Multiplexers, L298N Motor Drivers, PID Control Logic
Tools & IDEs:	Eclipse, Code::Blocks, Efficient Prompt Engineering

KEY TECHNICAL PROJECTS

ESP32 PID Line-Following Robot — **Embedded Systems Lab**

- Engineered an autonomous line-following robot utilizing an **ESP32 DevKit V1** microcontroller and N20 DC motors.
- Integrated a 16-channel TCRT5000 IR sensor array equipped with an external multiplexer for high-precision track detection.
- Implemented customized PID control algorithms for real-time error correction, ensuring stable tracking and high-speed navigation across complex tracks.

Telemedicine System Prototype — **Software Development Lab**

- Architected a functional telemedicine workflow prototype as part of rigorous university lab coursework.
- Applied core object-oriented principles and structured problem-solving methodologies to model modular user and data interactions.

EDUCATION & ACADEMIC CREDENTIALS

North South University, Bangladesh — Expected 2028
B.Sc. in Computer Science & Engineering (CSE) — Current CGPA: 2.42

Kabi Nazrul Govt. College, Dhaka — HSC — 2020 – 2022 | GPA: 5.00 / 5.00

Motijheel Govt. Boys' High School, Dhaka — SSC — 2011 – 2020 | GPA: 5.00 / 5.00

LEADERSHIP & EXTRACURRICULAR LEADERSHIP

KNGC English Club — **Vice President**
Dec 2021 – Aug 2022

- Directed club operations, coordinated team meetings, and organized academic public speaking events to foster effective communication skills among students.

MGBHS Green Club — **Executive of Management**
Dec 2018 – Jan 2020

- Managed logistics and student coordination for institutional sustainability initiatives and campus-wide events.

HONORS & ACHIEVEMENTS

- **Shadhinata Cup Football Tournament:** Gold Medalist, demonstrating high teamwork, endurance, and strategic coordination.
- **National Green Carnival:** Awarded Best Executive & 1st Runner-Up for exceptional leadership and event execution.
- **Business Fest '23:** Crowned Mobile Photography Champion among competitive peers.